# RAJALAKSHMI ENGINEERING COLLEGE

**(An Autonomous Institution)**
**Department of Computer Science and Engineering (Cyber Security)**

## CR23532 – CYBER AND DIGITAL FORENSICS
## CASE STUDY ASSIGNMENTS – II


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**Branch / Section :** B.E. CSE (Cyber Security)

**Year / Semester :** III Year / V Semester

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**Date of Submission :** 16 October 2026

# Overview of the Selected Scenarios

| Assignment | Questions | Scenario title | Deliverables included |
|---|---|---|---|
| 2(1) | 1–7 | The Placement Cell Résumé Database Export | Flowchart, evidence table, timeline, report structure |
| 2(2) | 8–14 | The Hidden Sync Agent on the Physics Lab Data-Logging PC | Inventory table, linkage matrix, ILookIX workflow, limitations |
| 2(3) | 15–22 | The Sports Department Selection-Trial Leak | Acquisition flowchart, imaging procedure, custody form, integrity explanation |
| 2(4) | 23–30 | The Altered Tender Comparison Sheet | Classification table, timeline, VSS workflow, validation checklist |
| 2(5) | 31–38 | The Unauthorised Grade-Revision Request | Evidence map, correlation matrix, challenges, final summary |

# 2(1): Suspected Unauthorized Access to a College Server

**Scenario: The Placement Cell Resume Database Export.** Around 01:20 a.m. on 12 October 2026, the Placement Cell's database server (PCELL-DB01) recorded a bulk export of about 2,400 student resume records, containing names, phone numbers, CGPA and email addresses, while the cell was closed. The Placement Officer has complained. The potentially relevant items are: the complaint, a staff workstation (PC-PLC-03), server access and database audit logs, a 64 GB USB drive found plugged into that workstation, and an exported file named resume_bulk.csv.

**Assumptions:** (a) Written authority from the Registrar covers PCELL-DB01, PC-PLC-03 and the USB drive only. (b) Database auditing and Windows security logging were switched on. (c) The workstation is a Windows 11 PC with NTFS. (d) The scenario is hypothetical; all times and names are illustrative and no tool was actually run.

## Q1. Forensic phases applied to this case

I treat the five phases as a pipeline in which each phase produces a defined output that the next phase consumes.

| Phase | What the team does in this case | Output handed forward |
|---|---|---|
| Identification | Fix the incident window (11 Oct 18:00 to 12 Oct 08:00). List every source that could show the export: DB audit trail, OS logs of the server, VPN/RDP logs, the workstation, the USB drive, resume_bulk.csv and the staff duty roster. Ask the server administrator to stop log rotation. | Scope note, source list, preservation requests |
| Collection | Photograph the workstation and USB in place. Take server logs as native exports by an authorised administrator. Image the workstation disk and USB behind a write blocker. Capture RAM only if the PC is still running and the decision is recorded. | Sealed exhibits, images, native logs, hash list, custody form |
| Examination | Verify hashes, then parse the copies: Windows event logs, USB device history in the registry, Recent/LNK items, browser downloads, NTFS $MFT entries for resume_bulk.csv, and the CSV's own metadata. | Extracted artefacts, each with its exhibit reference |
| Analysis | Place all events on one UTC timeline, link them through account name, session ID, source IP and file hash, and test competing explanations (a scheduled backup, a shared password, an authorised data pull by the Placement Officer). | Timeline, supported and unsupported statements |
| Presentation | Write the report and brief the review committee in plain language, stating what is proven, what is probable and what is unknown. | Signed report with appendices |

## Q2. Investigation flowchart

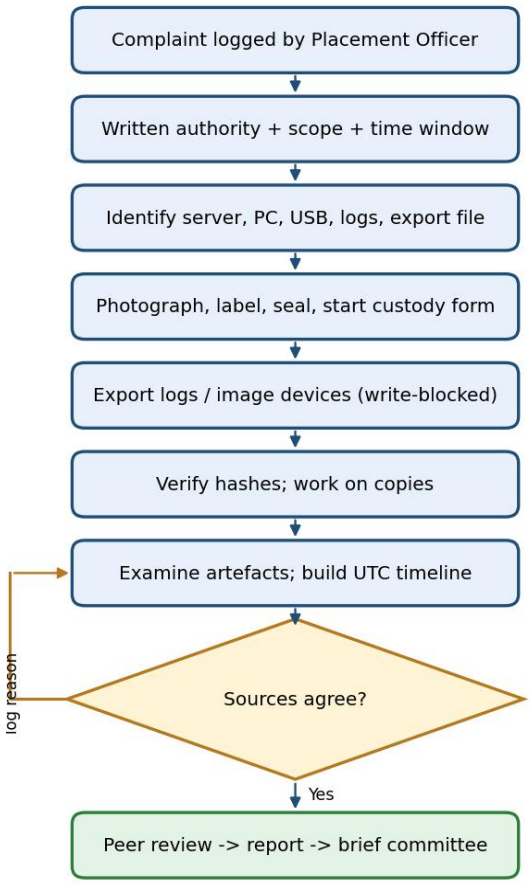



**Figure 1.1: Workflow from complaint to presentation, with a re-examination loop**

The loop at the decision point matters: if the server audit says one account exported the file but the workstation shows nobody logged in, the examiner does not pick the more convenient source. The conflict is logged and a targeted re-examination is performed, for example checking for remote sessions or a clock offset.

## Q3. Potential digital evidence and its investigative value

I use four value classes: **Direct** (records the act itself), **Corroborative** (supports a direct item), **Contextual** (frames the event) and **Lead** (suggests where to look next). The grading is provisional until the artifacts are examined.

| S.no | **Evidence item** | **Where it sits** | **What it may show** | **Value class** |
|---|---|---|---|---|
| 1 | Placement Officer's complaint | Registrar's file | Claimed time, data affected, who had access | Lead / contextual |
| 2 | Database audit trail | PCELL-DB01 | Account, query, rows exported, source host | Direct |
| 3 | Server Security event log (4624/4625 logons) | PCELL-DB01 | Logon type, source IP, session duration | Corroborative |
| 4 | PC-PLC-03 disk image | Placement Cell desk | Recent files, browser downloads, user profile activity | Direct if the export file is present |

| S.no | Evidence item | Where it sits | What it may show | Value class |
|---|---|---|---|---|
| 5 | USB drive (64 GB) | Plugged into PC-PLC-03 | Copy of resume_bulk.csv or deleted traces | Direct |
| 6 | resume_bulk.csv metadata and SHA-256 | Workstation or USB | File identity and whether the copies are identical | Corroborative |
| 7 | Registry USB history (USBSTOR) | Workstation image | Device serial and first/last connection time | Corroborative |
| 8 | Door-swipe log and duty roster | Admin office | Who was in the building | Contextual |

## Q4. Precautions in collection and preservation

- **Authority first.** Collect only what the written order covers; stop and seek an extension if the scope has to widen.
- **Record before touching.** Photograph screens, cables and ports; note power state, date/time on screen, serial numbers and visible labels.
- **Never browse the original.** Use a write blocker for the drive and USB, create bit-for-bit images and hash both source and image (SHA-256, with MD5 only as a secondary check).
- **Preserve logs natively.** Export server logs in their original format through an administrator, log the exact command or menu path, and hash the export.
- **Protect the exhibits.** Anti-static bags, tamper-evident seals, locked storage and a custody entry for every handover.
- **Clocks.** Record the system time against a reference time source on the day of collection.
- **Document everything, including failures,** such as tool errors, unreadable areas or any unavoidable change during live capture.

## Q5. Establishing a timeline of the access

The method is to gather each source's events, keep the raw timestamp and its time zone, compute a UTC column after measuring each clock's offset, and then join events using stable keys. The illustration below uses made-up values.

| Time (IST, illustrative) | Event | Source | Limits of the evidence |
|---|---|---|---|
| 12 Oct 00:58 | VPN session opens for account plc_asst2 | VPN log | Account, not person |
| 12 Oct 01:03 | RDP logon to PCELL-DB01 (logon type 10) from VPN pool address | Server Security log | Confirms the session, not who typed |
| 12 Oct 01:20 | Bulk SELECT and export, about 2,400 rows | DB audit trail | Shows the action took place |
| 12 Oct 01:24 | resume_bulk.csv written on PC-PLC-03 | NTFS $MFT, Prefetch | Timestamps may be changed by copying |
| 12 Oct 01:26 | USB serial connected to PC-PLC-03 | Registry / setupapi log | Connection is not a copy |
| 12 Oct 01:27 | Identical CSV found on USB (same SHA-256) | USB image | Strong for transfer; no operator identity |
| 12 Oct 08:10 | Officer notices the export alert | Complaint | Discovery time |

A hash match between the CSV on the workstation and the one on the USB shows the files are the same. It takes the door log, local logon and VPN records together to say anything about who might have been present. Any single timestamp is just a lead.

## Q6. Final forensic report and presenting it to a non-technical authority

| Report section | Content |
|---|---|
| 1. Cover and summary | Case number, examiner, date, a half-page summary in everyday language |
| 2. Authority and scope | Who authorized, what devices and time window, exclusions |
| 3. Exhibits | Exhibit IDs, descriptions, condition at receipt, seal numbers |
| 4. Methods and tools | Imaging and parsing tools with versions, hash algorithm, settings |
| 5. Observations | Numbered facts, each pointing to its source artefact |
| 6. Timeline | Table in IST and UTC with a confidence note for each row |
| 7. Analysis | Explanations considered, including those that do not implicate anyone |
| 8. Conclusions and limits | What is established, probable, and unknown |
| 9. Appendices | Hash manifest, custody forms, log excerpts, glossary |

For the committee I would use a one-page timeline, define two terms (hash = a digital fingerprint of a file; audit log = the server's own diary), and keep fact and opinion visibly separate, for example with a column labelled "Observed" and another labelled "Interpretation". I would avoid naming a culprit unless independent evidence supports it.

## Q7. Legal and ethical considerations in a college environment

- **Authority and policy.** Written approval from the competent college authority and compliance with the IT-usage policy staff and students agreed to.
- **Law.** The IT Act 2000 (offences such as unauthorized access), the Digital Personal Data Protection Act 2023 (the resume data is personal data) and the Bharatiya Sakshya Adhiniyam 2023, which sets the certificate requirement for electronic records. The college legal adviser should confirm how these apply.
- **Minimization.** Examiners see only the data needed for the allegation; unrelated student records stay masked in reports.
- **Fairness.** Report evidence that clears the suspect as well as evidence against, and give the person a chance to explain.
- **Confidentiality and independence.** Restricted access to evidence and no examiner with a conflict of interest.

# 2(2): Linking a Suspicious File to a User Using File-System Evidence

**Scenario: The Hidden Sync Agent on the Physics Lab Data-Logging PC.** PHY-DL02 collects readings from lab instruments and is used by a lab assistant, two research scholars and, occasionally, a guest account. A technician finds sync_agent.exe inside a hidden folder (C:\ProgramData\.cache_tmp), next to two encrypted files, calib_backup.vault and notes_q3.vault. Browser traces also show visits to a file-sharing site. The question is which account, if any, created and used these files.

**Assumptions:** Windows 11, NTFS, a verified disk image can be made, ILookIX is available, authorization is on file, and no account holder is presumed responsible. All values below are illustrative.

## Q8. Hardware and software to document before analysis

| Area | Details to record | Reason |
|---|---|---|
| Machine | Make, model, serial, asset tag, BIOS/UEFI version, photographs, attached instrument interfaces | Identity of the exhibit |
| Storage | Drive model and serial, capacity, partition layout, sector size, BitLocker or other encryption state | Selects imaging method; flags limits |
| Memory and power | RAM size, whether on or off, boot order, removable media attached | Volatile evidence decision |
| Operating system | Edition and build, install date, time zone, patch level | Controls which artifacts exist |
| File system | NTFS volumes, mount points, $MFT and $UsnJrnl presence | Metadata availability |
| Accounts | Local usernames, SIDs, group membership, last-logon times, password policy | Needed to attribute artifacts |
| Software | Installed applications, instrument drivers, antivirus, backup/sync clients, scheduled tasks, tool versions used by the examiner | Separates normal software from the suspect file |
| Network | Hostname, IP/MAC, domain or workgroup, proxy settings | Cross-system correlation |

Most of this comes from scene notes and the read-only image; the original machine is not booted just to complete a form.

## Q9. How storage, operating systems, file systems and metadata help locate evidence

A storage device holds active files, deleted entries and slack or unallocated space. The operating system adds its own records: registry hives, event logs, Prefetch, Jump Lists and user profiles. NTFS contributes the Master File Table, which stores for each file its name, parent folder, size, owner SID and four timestamps (created, modified, MFT-changed, accessed), and the $UsnJrnl, which logs changes. For sync_agent.exe the combined view answers practical questions: where it sits, when it first appeared, which profile owns it, whether it was renamed, and whether anything executed it. Each of these answers is a clue, and clues become evidence only when they agree.

## Q10. Hidden and encrypted files: effect on the investigation and handling

| Issue | Why it complicates the case | Handling |
|---|---|---|
| Hidden attribute / dot-folder | Item does not show in normal views, but this alone is not proof of wrongdoing | Examine through the image; keep attributes unchanged; record the full path |

| Issue | Why it complicates the case | Handling |
|---|---|---|
| Alternate data streams | Content may be attached to a harmless-looking file | Enumerate streams in the tool; export separately with hashes |
| Wrong extension | A file named .txt may be an executable | Check the header signature, not the name |
| Encrypted containers (.vault) | Contents unreadable without a key; the file's existence, size and access times still tell something | Hash the original, never alter it, use a key only if lawfully supplied, and decrypt a copy |
| Full-disk encryption | May block imaging of a powered-off disk | Record the state; consider a live acquisition only under an approved plan |

- State openly that encrypted content was not reviewed if no key is obtained; do not guess what is inside.
- Keep personal or unrelated research data out of the report even though it remains in the image.

## Q11. Evidence-linkage table

The rows show how I would test a link between sync_agent.exe and one account. Everything in the third column is a placeholder to be replaced with verified findings.

| Artefact | Where found | Illustrative value | Supports | Strength |
|---|---|---|---|---|
| File path | $MFT | C:\ProgramData\.cache_tmp\sync_agent.exe | Location only (ProgramData is machine-wide) | Weak alone |
| Created time | $MFT $STANDARD_INFORMATION and $FILE_NAME | 09 Oct 16:42 in both | Consistent creation; mismatch would suggest timestamp tampering | Moderate |
| Owner SID | NTFS security | SID ending -1004 (scholar1) | File created under that account's context | Moderate |
| Browser download | scholar1 Chrome profile | URL for sync_agent.exe, 16:40 | Source of the file | Moderate |
| Prefetch | C:\Windows\Prefetch | SYNC_AGENT.EXE run count 3, last 16:45 | Execution happened | Strong for execution, none for person |
| Scheduled task | Task Scheduler | Task "LabSync" runs the file at logon | Persistence or approved job | Needs creator check |
| Logon event | Security.evtx | scholar1 interactive logon 16:30 to 17:20 | Account was active | Contextual |
| LNK / Jump List | scholar1 profile | Shortcut to calib_backup.vault | User interacted with the container | Moderate |

| Artefact | Where found | Illustrative value | Supports | Strength |
|---|---|---|---|---|
| Hash | ILookIX hash set | SHA-256 compared with the vendor's copy | Identifies the program | Strong for identity of file |

## Q12. ILookIX workflow

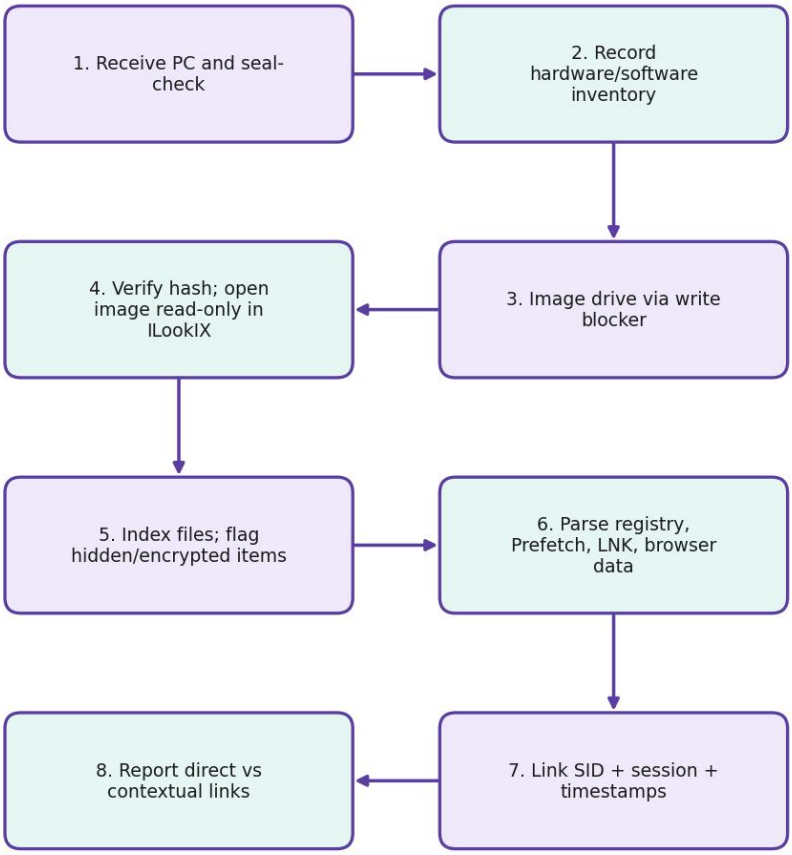



**Figure 2.1: ILookIX analysis workflow for PHY-DL02**

1. **Receive and verify.** Check seals; image the drive with a write blocker; verify and record the image hash.
2. **Create the case.** Name the examiner, record the tool version and the clock offset of the examination computer.
3. **Import read-only.** Add the verified image; confirm that partitions and NTFS are recognized; note any warnings.
4. **Index and filter.** Build the file index; list hidden, system and encrypted items; run a signature check against extensions.
5. **Targeted parsing.** Registry hives (accounts, USB, run keys), event logs, Prefetch, LNK/Jump Lists, browser history and downloads.
6. **Timeline and linkage.** Merge artefact times into one timeline and tie them to SIDs and logon sessions.
7. **Export and report.** Export artifacts with path, hash and source image; generate the report; list what the tool could not parse.

The expected outputs are an import log, an account inventory, an executable profile (hash, path, signature), a linkage matrix, an export manifest and an analysis log. These are intended outputs, not results of a tool run.

## Q13. Direct links versus supporting or contextual information

| | **Directly links a user to the activity** | **Only supports or gives context** |
|---|---|---|
| Definition | Artefact that records the act and carries an account identifier | Artefact that fits the story but has other possible explanations |
| Examples here | Process-execution audit with the SID of scholar1; user-profile LNK for calib_backup.vault; download entry in scholar1's own browser profile | File simply sitting in Program Data; USB serial in the registry; lab timetable; antivirus scan entry |
| Caution | Even these link an account, not a human at the keyboard | Never enough on their own |

## Q14. Limitations and alternative explanations

- **Shared environment.** The guest and scholars use one machine; an unlocked session or shared password could explain the activity.
- **Legitimate software.** An instrument vendor's sync utility, or a backup tool, may look like sync_agent.exe; the hash and vendor signature decide.
- **Automated creation.** Scheduled tasks and updaters can create files without anyone clicking.
- **Time problems.** Clock errors, file copies and restored backups can shift timestamps.
- **Missing or unreadable data.** Encrypted containers stay unknown; cleared logs or disabled auditing reduce what can be shown.
- **Remote access.** Another person could have worked through a remote-desktop session under the same account.

## Conclusion

If browser, MFT, Prefetch and logon evidence all point to scholar1's profile, the report can say that the activity is associated with that account. It cannot say that the scholar personally ran or hid the file until shared use, remote access and automation are ruled out. If the sync agent turns out to be an approved vendor tool, the matter closes as a misunderstanding, and that outcome is as legitimate as any other.

# 2(3): Forensic Imaging and Safekeeping of a Suspect Storage Device

**Scenario: The Sports Department Selection-Trial Leak.** Shortlist scores for a university team trial appeared on social media before they were officially released. A laptop (SPT-LP05) from the sports office was seized from a staff desk. It was closed and in sleep mode, with a 500 GB external hard disk attached by USB. The task is to collect both devices without changing them and to image them for analysis.

**Assumptions:** Written seizure authority exists; a write blocker, a bootable acquisition environment with IXImager (or a documented equivalent), and secure storage are available. The workflow describes the intended procedure; no imaging was actually done. Names, times and IDs are illustrative.

## Q15. Collecting and transporting the laptop and USB device

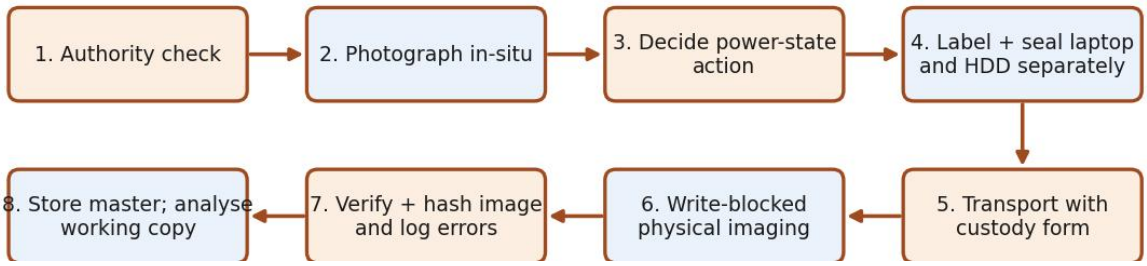



**Figure 3.1: Collection-to-analysis flow for the laptop and external HDD**

1. **Authority.** Check the seizure order, note the case number and the people authorized to act.
2. **Scene record.** Photograph the desk, laptop (lid closed), HDD, cables, charger, ports and any notes on screen.
3. **Power-state decision.** A laptop in sleep is not off. Record this, tell the lead examiner, and follow the plan: either a documented live capture (memory, encryption keys if lawful) or a controlled shutdown. Do not open files or let it connect to Wi-Fi; if the lid must be opened to check, note it.
4. **Isolate the network.** Use the approved method such as a shielded bag or disabling radios, and record what was done.
5. **Disconnect carefully.** Note the port the HDD used, remove it, and bag it as its own exhibit.
6. **Identify and label.** Record make, model and serial numbers; assign SPT-01 (laptop), SPT-02 (HDD), SPT-C (charger); apply tamper-evident seals.
7. **Pack and carry.** Padded anti-static packaging, no stacking of devices, kept dry and away from magnets and heat.
8. **Custody.** Complete a custody entry at each handover; deposit in a secure, access-logged evidence locker.

## Q16. Purpose of physical acquisition, and why analysis uses an image

A physical acquisition copies every addressable sector, so the image holds the partition table, file-system structures, active files, deleted-file remnants and unallocated space, not just what Windows Explorer shows. This is what makes deleted score sheets or earlier file versions recoverable.

Analyzing an image rather than the original protects the original from accidental writes (an operating system will modify timestamps and create files simply by mounting a disk), allows the examination to be repeated and checked by another examiner, and means a failing disk is read as few times as

possible. A caution: SSD TRIM, hardware encryption and unreadable sectors can still limit what any image contains, so the report states exactly what was captured.

## Q17. IXImager imaging workflow

IXImager is a legacy, Linux-based imaging tool linked to the ILook family. Before relying on it I would confirm that the available version supports the devices and the intended output format; if not, a validated equivalent is documented in the report.

| Stage | Actions | Records kept |
|---|---|---|
| Source identification | Boot the acquisition environment; list attached disks; match model, serial and capacity to the exhibit label | Source description, device path, scene inventory comparison |
| Write protection | Attach through a tested hardware blocker (or a software read-only mode where a hardware one is impossible); confirm the OS did not mount the source | Blocker model, test result, mount status |
| Destination | Prepare clean, formatted, adequately sized storage with its own case folder | Destination serial, free space |
| Imaging | Choose physical (whole-disk) acquisition and a supported evidence format; set error handling for bad sectors | Tool version, options, start/end time, error list |
| Verification | Run the tool's verification and independently re-hash the completed image | Source hash (if available), image hash, result |
| Hashing and documentation | Hash each segment and the container; save logs; photograph the setup | Hash manifest, acquisition log |
| Safekeeping | Seal originals; lock the master image; make a working copy | Custody entries, copy hashes |

The laptop and the external HDD are imaged separately because each has its own serial number, hash and history; an image of one says nothing about the other.

## Q18. Using hash values to demonstrate integrity

A cryptographic hash such as SHA-256 turns any amount of data into a fixed-size value, and changing even a single bit changes the value completely. I compute the hash of the source as it is read, the hash of the image straight after acquisition, and then again whenever the image is copied or opened for analysis. Identical values show the data has not changed between those points.

- The comparison must be like with like: a compressed container hash will differ from the raw-data hash inside it, and both can be right if labelled.
- If bad sectors made some reads fail, the report states that the match applies to the data actually read, and lists unreadable address ranges.
- Any mismatch is investigated and explained, never ignored.

## Q19. The ASB container

ASB is an evidence-image format associated with IXImager and ILook, intended to store the acquired data together with descriptive and authentication information so that compatible tools can open it. I would only rely on it after checking the installed version's support, and I would keep a record of any conversion to another format. The package should travel with:

- Case and exhibit numbers, source serial number and description.
- Examiner name, acquisition date, time and time zone.

- IXImager version, options, segment list and image size.
- Source and image hashes, and verification results.
- Bad-sector log, condition notes and the custody record.

## Q20. Data carving

Data carving scans raw sectors for the start and end signatures of known file types (for example the FF D8 FF header of a JPEG or the %PDF header) and rebuilds files without using the file-system directory. It is needed when the directory entries are deleted, the partition has been reformatted or the file system is damaged. In the sports case it might recover a deleted score spreadsheet or photos of a printed result sheet.

Limits: carved files often lose names, paths and timestamps, fragmented files may be incomplete or joined wrongly, and false positives occur. I would carve only on a working copy, log offsets and settings, open each result to test it, hash it and label it "carved, origin unverified".

## Q21. Date and time problems

| Problem | Example in this case | Documentation approach |
|---|---|---|
| Clock error | Laptop BIOS clock is 7 minutes fast | Compare against a reference on seizure day; record the offset |
| Time zone | Cloud posting logs in UTC, laptop in IST | Keep the raw value and its zone; add a UTC column |
| Timestamp meaning | Copying the file to the HDD resets its created time | Note what each NTFS field means; compare $STANDARD_INFORMATION and $FILE_NAME |
| Precision | FAT-formatted HDD stores 2-second modified times | State the resolution |
| Tampering | User changed the system date | Look for time-change events and corroborate with external records |

## Q22. Chain-of-custody table (laptop SPT-01)

Mandatory fields for every row: exhibit ID, description and serial number, released by, received by, date/time with zone, purpose, seal number and condition, signatures. Entries below are illustrative.

| Date / time (IST) | Action | From | To | Seal / condition |
|---|---|---|---|---|
| 13 Oct 2026 10:05 | Seized at sports office | Sports officer (witness) | Collector Ravi | Seal SP-201 applied; lid closed, no damage |
| 13 Oct 2026 10:40 | Transport to lab | Collector Ravi | Courier Meena | SP-201 intact |
| 13 Oct 2026 11:15 | Lab intake | Courier Meena | Evidence custodian | SP-201 intact; locker L-6 |
| 14 Oct 2026 09:30 | Issued for imaging | Evidence custodian | Examiner Kiran | SP-201 broken in front of witness |
| 14 Oct 2026 16:10 | Imaging complete, returned | Examiner Kiran | Evidence custodian | New seal SP-202; image hash recorded |

## Evidence-integrity summary

The originals are never examined; the master image is stored sealed and read-only; and all analysis runs on a verified working copy whose hash is rechecked before and after use. Anything that could not be read, or any step that departed from procedure, is written into the report rather than left out.

# 2(4): Recovery of Deleted Evidence from Volume Shadow Copies

**Scenario: The Altered Tender Comparison Sheet.** The Estate Office compares vendor quotes for a 125 kVA generator in a spreadsheet, Tender_Comparison_Gen125.xlsx. After the committee reviewed it, the lowest bid figure appears to have been changed and the file was later removed from the shared folder. A verified forensic image of the Estate Office desktop (Windows 11) is available, and System Protection was on, so Volume Shadow Copies (VSS) may hold an earlier version.

**Assumptions:** The image hash was verified; relevant event logs, the NTFS $MFT and $UsnJrnl, and Recycle Bin records are present; a cloud-sync client is installed on the desktop. Dates and names are illustrative.

## Q23. Selecting and categorizing evidence before detailed analysis

1. **Define the question.** Which version of the tender sheet was approved, what changed, when, and was it deleted or moved?
2. **Confirm the baseline.** Check image identity and hash, the volumes present, system clock and whether shadow copies exist.
3. **Select by relevance.** Target file name patterns, the tender reference number, the Estate Office folders, the incident dates, Office recent-file lists, Recycle Bin entries, $MFT/$UsnJrnl records and sync logs.
4. **Reduce noise.** Exclude known operating-system and application files using a trusted hash set, and keep a record of the filter and how many files were excluded.
5. **Categorise.** Sort what remains into the classes of Q24 so that each item has a clear role.
6. **Minimise.** The whole image stays preserved, but unrelated documents are not opened or reproduced.

## Q24. Classification scheme

| Class | Examples in this case | How it is treated |
|---|---|---|
| User-created | Tender_Comparison_Gen125.xlsx, vendor quote PDFs, approval note | Primary content; recover and compare versions |
| System / application | Windows event logs, Office AutoRecover files, $MFT, $UsnJrnl | Explain timing and behaviour; parser output is validated |
| Potentially suspicious | Renamed copy such as tender_final2.tmp, a cleanup script, wrong extension | Needs checking; "suspicious" means "examine further", not "guilty" |
| Deleted / recovered | Recycle Bin $I/$R pair, VSS version, carved fragment | Record origin (snapshot ID, offset); validate before use |
| Supporting / contextual | Sync-client logs, email approving the bid, logon events, browser history | Corroborates timing and account activity |

## Q25. Event analysis

Event analysis puts records from different artifacts on one clock to show a sequence. For the tender sheet, $UsnJrnl shows file changes (data overwrite, rename, delete), Office recent-file data shows when the workbook was opened, Windows logs show who was logged on, the sync log shows uploads, and VSS metadata shows snapshot times. If every source places the modification between the same two snapshot times, confidence rises. If sources disagree, the discrepancy itself becomes a finding to explain, usually through a clock offset or a different meaning of the timestamp. Sequence does not show intent: a save that rewrote the file and a deliberate edit can look identical in the journal.

## Q26. Cloud Analysis and Lead Analysis

| Technique | Purpose | Questions it helps answer | Caution |
|---|---|---|---|
| Cloud Analysis | Examine authorised cloud storage, email or collaboration audit data | Was the sheet uploaded, shared, renamed or restored? Which account and session? Does the cloud hold a version history that the PC lacks? | Retention differs by provider; account activity is not a person |
| Lead Analysis | Follow one clue to other sources | Where else does the tender reference or the file's SHA-256 appear? Who received it by email? Which USB or folder shows the same name? | A lead is a hypothesis until corroborated |

## Q27. Volume Shadow Copy analysis

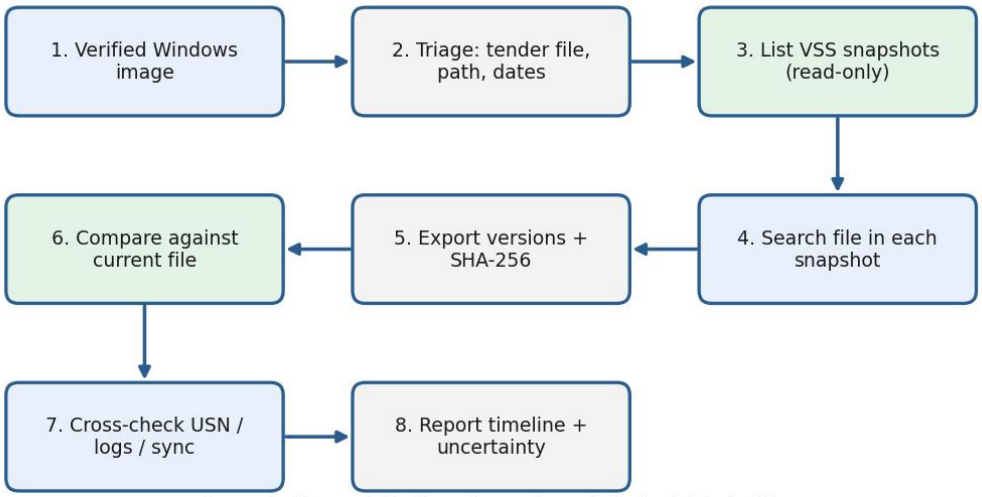



**Figure 4.1: Recovering and validating earlier versions from VSS**

VSS stores point-in-time snapshots of a volume, so a file edited or removed after a snapshot can still be found in its earlier state. Steps: identify the snapshots in the image, record their IDs and creation times, mount or browse them read-only, locate the tender file in each, export every version with its path and snapshot ID, hash each export, and compare content and metadata with the current state (or its absence). Because a snapshot only captures the moment it was taken, versions made between snapshots may not exist, and old snapshots may already have been purged.

## Q28. Hypothetical file life-cycle timeline

| Date and time (IST, illustrative) | Event | Evidence source |
|---|---|---|
| 09 Oct 10:05 | Sheet created with three vendor bids | $MFT creation time; document properties |
| 10 Oct 14:30 | Committee review; lowest bid recorded as Rs 18.4 lakh | Office revision data; email approval |
| 10 Oct 22:00 | Scheduled restore point (VSS-B) taken | VSS metadata |
| 11 Oct 20:12 | Lowest-bid cell changed to a higher figure and saved | Office metadata; $UsnJrnl data-overwrite event |
| 11 Oct 20:20 | Copy uploaded to the sync folder | Sync client log; cloud object ID |
| 12 Oct 08:55 | Sheet deleted from the shared folder | $UsnJrnl delete reason; Recycle Bin $I file |

| Date and time (IST, illustrative) | Event | Evidence source |
|---|---|---|
| 15 Oct 11:30 | Version in VSS-B exported read-only and hashed | Extraction log; custody entry |

This sequence would be read cautiously: the USN entry might come from an Office temp-file save, the deletion might be a move, and the sync client could have changed the file without any user action.

## Q29. Validating the recovered artefact

- Verify the source image hash before and after extraction.
- Record snapshot ID, volume, snapshot time, original path and extraction tool version.
- Give the exported file an exhibit ID and SHA-256; re-extract once to confirm the same hash is obtained.
- Check the file type by signature, open it in two different viewers, and confirm it opens without repair.
- Compare content with the current file and any other copies; describe exactly which cells or fields differ.
- Cross-check timestamps against logs and sync records, keeping raw and normalized times.
- Note anything incomplete, corrupted or missing, and what the snapshot can and cannot establish.

## Q30. Sources of error and alternative explanations

- **Clock and time zone errors** can reorder events; one tool may show UTC and another local time.
- **Low timestamp resolution or timestamp copying** can hide the true order of rapid edits.
- **Application behaviour.** Office writes a temp file and renames it during every save, which looks like delete-and-create in the journal.
- **Sync or backup software** may overwrite, restore or remove files without user action.
- **Authorized editors.** Another office employee, or an approved correction, may explain the change.
- **Snapshot gaps.** Edits between snapshots are invisible; snapshots may have been deleted by cleanup or low disk space.
- **Incomplete recovery.** Overwritten data or missing metadata may prevent a full restore.
- **Deletion is not concealment.** A missing file shows the file is absent, not why.

## Closing validation statement

If VSS-B is validated, the report can say that a version of the tender sheet with the lower bid existed on 10 October 22:00. If USN, Office and sync records agree, it can also say the later change and removal are corroborated. It cannot name the person responsible without separate login or process evidence. Deliverable included: classification table (Q24), timeline (Q28), VSS workflow (Figure 4.1) and validation checklist (Q29).

# 2(5): Multi-Source Investigation of a Suspected Cyber Incident

**Scenario: The Unauthorized Grade-Revision Request.** At 00:47 a.m. on 14 October 2026, a grade-revision request (GR-2291) for a failed subject was submitted through the college ERP using a third-year student's account. The student says they did not submit it. With consent and authority, investigators may examine the student's Windows laptop, browser data, email, a messaging app, the student's phone and the ERP's cloud audit records.

**Assumptions:** Consent from the student and written authority from the college are on file; the ERP vendor will supply audit exports through a formal request; clocks across sources may differ. Times and identifiers are illustrative, and nobody is presumed responsible.

## Q31. Evidence from Windows and other operating systems

- **Windows laptop:** logon events (4624/4634), Windows Hello or password sign-ins, user profile creation, registry (recent documents, USB, network profiles), Prefetch, LNK and Jump Lists, event logs, scheduled tasks, installed remote-access tools, system time zone and NTFS metadata.
- **Other systems:** if a Linux or macOS device was used, auth logs, shell history, browser profiles and application support folders; on a phone OS, see Q34.
- Observe three separate layers: signed in to Windows, browser profile open, and valid ERP session. Each is a different fact.

## Q32. Browser artefacts and careful interpretation

| Artefact | What it can reveal | Why interpret with care |
|---|---|---|
| History and visit times | ERP pages visited, in what order | Prefetching, sync from another device or a shared profile can add visits |
| Cache and form data | Draft text of the request | May be stale, partial or from an earlier visit |
| Cookies and session tokens | Whether an authenticated session existed | A stolen token works from anywhere |
| Downloads | Files obtained before the request | A download is not an upload |
| Saved passwords and extensions | Credential exposure, injected scripts | Presence does not show misuse |

Private browsing, history clearing and browser updates all change what survives, so every browser clue is checked against the ERP's own logs.

## Q33. Email and messaging evidence and correlation

- **Email:** full headers (Received chain, Message-ID, authentication results), sender and recipient addresses, security alerts such as "new sign-in", password-reset mails, OTP messages, attachments.
- **Messaging:** exported chats with sender IDs, timestamps, attachments and links; delivery or read states; notification records on the phone.
- **Correlation:** match a reset email with the ERP's reset event; match the OTP delivery time with the login attempt; compare the attachment hash with any uploaded file; match message times with browser visits.
- Preserve original exports. A screenshot shows only what was on screen and omits headers.

## Q34. Mobile phone examination

- Record model, OS version, IMEI, lock state, SIM/eSIM, battery and network status; photograph the screen.
- Confirm lawful consent for this phone and limit the scope to the incident.
- Prevent remote wipe or sync changes by an approved isolation method, and document it.
- Choose a validated acquisition type (logical, file-system or physical) suited to the model and state which one was achieved.
- **Challenges:** strong device encryption, locked bootloaders, app sandboxing, disappearing messages, cloud-only data, frequent OS updates, many different models, and notifications that sync from other devices.
- Do not open personal apps on the original phone or browse beyond the authorised data.

## Q35. Internet and cloud evidence sources

| Source | Contents | Notes |
|---|---|---|
| ERP cloud audit | Logins, source IP, user agent, session ID, request ID GR-2291, MFA events | Strongest server-side record; request through proper channel |
| Identity provider / SSO | Sign-in history, device, MFA result | Shows how the login happened |
| Email provider | Sign-in alerts, reset records, message trace | Needs account authority |
| Campus network | DHCP, proxy and firewall logs, VPN logs | Contextual; NAT hides individual users |
| Telecom / app providers | OTP delivery records, app account data | Legal process normally required |

IP addresses and user-agent strings are context only. The report records the retention period of each source, since some logs disappear within weeks.

## Q36. Multi-source evidence map and cross-source correlation table

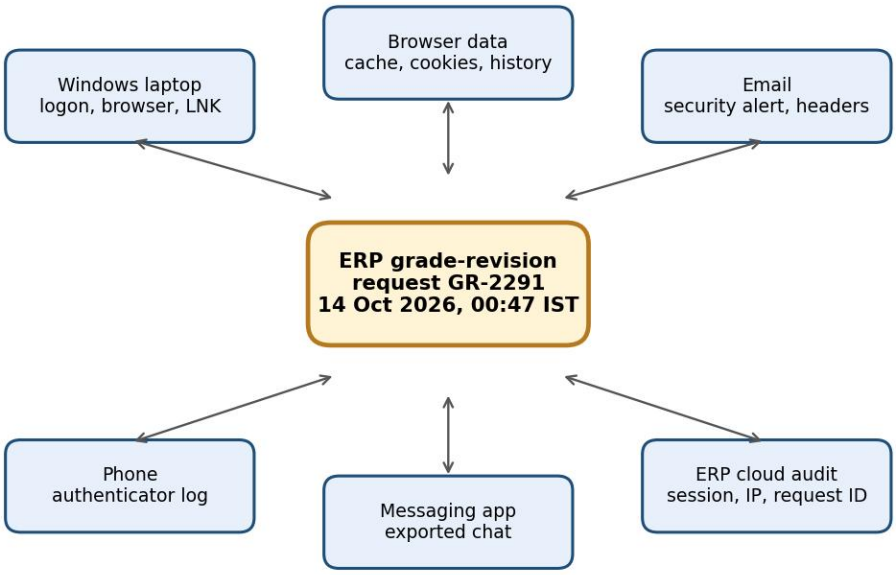



**Figure 5.1: Evidence sources converging on request GR-2291**

| Source | Hypothetical artefact | Correlation key | Meaning | Caution |
|---|---|---|---|---|
| Windows laptop | Student account logon 00:31, unlock at 00:40 | Account SID, time | Account active on laptop before the request | Session may have been used by someone else |
| Browser | ERP login page and grade-revision form in cache, 00:42 | URL, profile, time | Form opened on this laptop | Cache may predate the submission |
| Email | "New sign-in to ERP" alert, 00:44 | Message-ID, sender | A login was recognized by the provider | Verify headers; could be phishing |
| Phone | Authenticator approval entry, 00:45 | Transaction ID | An MFA prompt was approved | Does not show who tapped it |
| Messaging app | Chat at 00:58 saying "request sent" from the student's account | Message time, contact | Possible knowledge of the request | Account could be impersonated |
| ERP cloud | Request GR-2291 POST, session S-77, IP from campus hostel range | Request ID, session ID | Server accepted the request | IP is not a person |
| Network | Hostel DHCP lease and NAT mapping | IP and time | Plausible location | Many users share the gateway |

## Q37. Challenges when evidence is spread across devices and the cloud

1. **Time inconsistency:** devices and services use different zones and clocks.
2. **Legal and ownership boundaries:** personal phones, provider data and jurisdiction need separate authority.
3. **Short retention:** cloud logs, chats and sessions can disappear quickly.
4. **Encryption and locked devices** limit extraction.
5. **Identity ambiguity:** accounts, IPs and MFA do not prove the person.
6. **Format and tool variety:** each app and OS stores data differently and changes with updates.
7. **Sync side-effects:** one action may appear on many devices, creating duplicate or misleading traces.
8. **Volume and privacy:** large quantities of unrelated personal data must not be reviewed.
9. **Chain of custody:** exports from dashboards and screenshots are hard to authenticate later.

## Q38. Final investigative summary

| Tier | Possible conclusion for GR-2291 | What would strengthen it |
|---|---|---|
| Direct | A verified ERP audit shows the student's account submitted GR-2291 at 00:47 IST in session S-77 | Original export from the vendor, request provenance, hash of the export |
| Corroborating | Laptop logon, cached form, sign-in email, MFA approval and chat message fall within the same few minutes | Time normalisation, authentic headers, device-to-account ownership |
| Unresolved | Who was at the laptop; whether the password or session was stolen; whether the MFA approval was knowing or tricked | Interviews, other login history, session-revocation records, lawful further records |

My provisional position is that the evidence may show the student's **account** was used for the request. It does not yet show that the student **personally** made it. The student is treated as a possible victim of account misuse as much as a possible participant. Recommended steps are to preserve the remaining logs, reset credentials, review MFA and session settings, and tell the student that the investigation is continuing.